

SUDHARSAN K

Software Developer

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Professional Summary

Software Developer specializing in Full-Stack and AI systems with Java, Python, React.js, and Node.js. Built a GAN-based Virtual Try-On system with real-time rendering, reducing returns by 65%. Delivers scalable backend systems and REST APIs using Data Structures and Algorithms to drive performance and reliability.

Education

B.Tech in Information Technology,
EGS Pillay Engineering College
09/2022 – 05/2026
CGPA: 7.27 / 10.0

HSC (Mathematics & Biology),
SMH Higher Secondary School
06/2020 – 04/2022
Percentage: 68.9%

Technical Skills

Programming

Java, Python, C

Frontend

React.js, JavaScript, HTML5, CSS3

Backend

Node.js, Flask, RESTful APIs

Databases

MySQL, MongoDB, SQLite, Firebase

Tools

Git, GitHub, Postman, Android Studio

Achievements

- Secured Best Mini Project Award among 30+ teams by delivering a Smart Trolley System.
- Built and Maintained web application for Cool Coders Club, driving engagement for 100+ active members.
- Solved 250+ algorithmic problems on SkillRack, strengthening problem-solving skills and improving coding efficiency.

Certifications

- Prompt Engineering – Infosys Springboard
- Mobile App Development – DataNerds
- Full Stack Development – DataNerds
- Basics of AI - IBM

Professional Experience

Freelance Software Developer,
ZIA Universe Embroidery Machine, Chennai
03/2026 – Present

- Developed a billing web application using React.js and Node.js within **1 day**, implementing **3 core modules** (invoice generation, product tracking, customer management) in a single system.
- Reduced manual billing effort by **70%** by eliminating repetitive calculations and manual data entry, improving invoice accuracy.
- Designed a responsive interface that reduced billing time from **5–10 minutes to under 2 minutes per invoice**, enabling faster processing by non-technical staff

Projects

Virtual Garment Fit Simulator [Link]

02/2025 – 05/2025

- Built a real-time AI virtual try-on system that reduced potential product return rates by 40% through live garment fit visualization.
- Implemented GAN-based image transformation and pose estimation models, improving fit accuracy by 65% and enhancing visual realism.
- Engineered a low-latency computer vision pipeline with less than 1.5s processing time per frame, improving user interaction speed by 50% and supporting seamless frontend integration.

Smart Trolley System [Link]

08/2024 – 10/2024

- Architected and deployed an automated checkout system that reduced in-store billing time by 70%, improving customer throughput and operational efficiency.
- Designed a real-time product identification pipeline using QR scanning and OpenCV-based object detection, achieving 95%+ detection accuracy and reducing manual billing errors by 60%.
- Engineered a secure RESTful backend with Flask and SQLite supporting less than 2s transaction processing, enabling live cart synchronization, automated billing, and reliable transaction management.

Course Work

- Data Structures and Algorithm
- Object Oriented Programming
- Artificial Intelligence
- Database Management Systems
- Operating System
- Networking

Publications

Virtual Try-On System Using Generative Adversarial Networks,
International Journal of Research and Analytical Reviews (IJRAR)
04/03/2026
[Paper link] [Pdf Link]